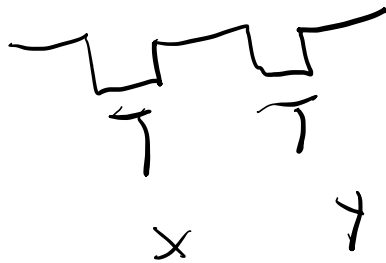
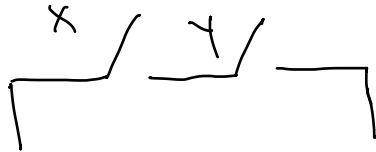


CMOS Design Styles

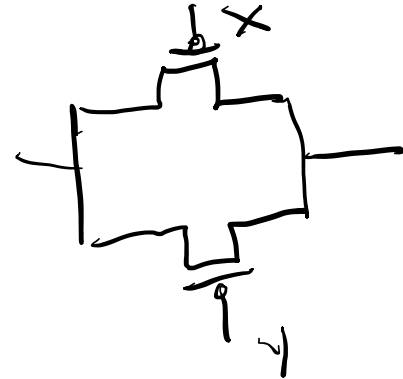
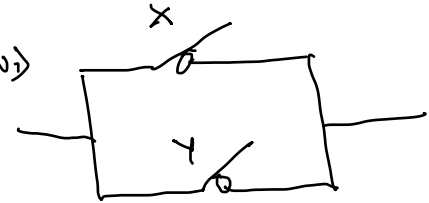
Ranga Vemuri

Pass Transistor Logic

AND

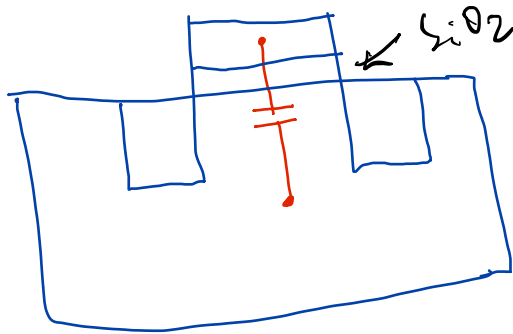


NAND

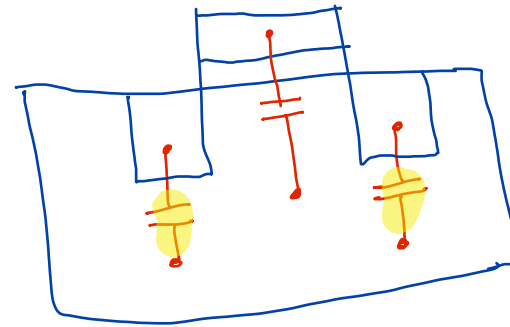


MOS Capacitances

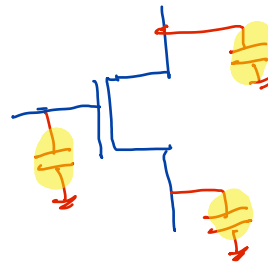
Gate Capacitance



Diffusion Capacitance



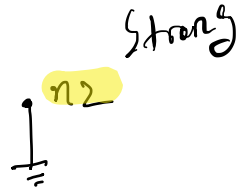
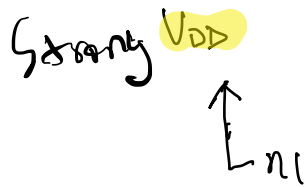
Circuit Representation



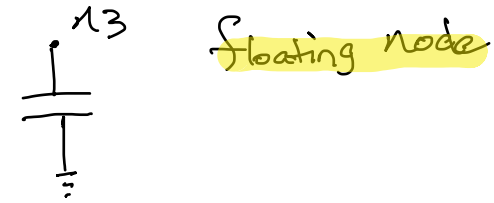
Parasitic
 C_{eq}

Strong and Weak Signals

Strong signals



Weak signals



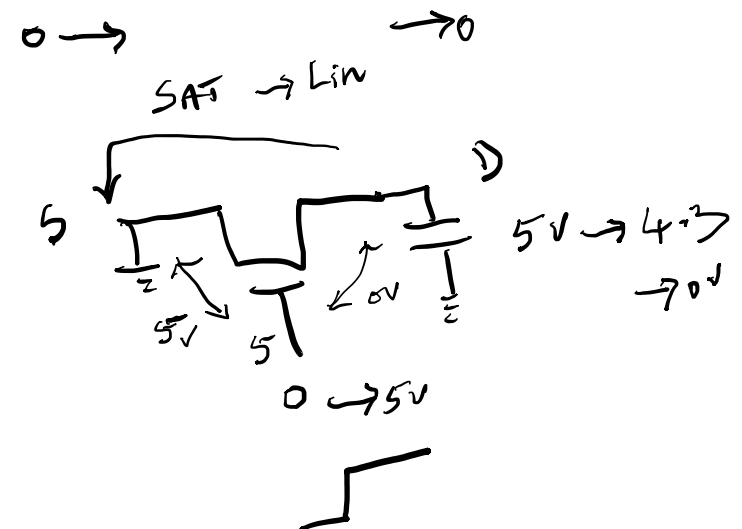
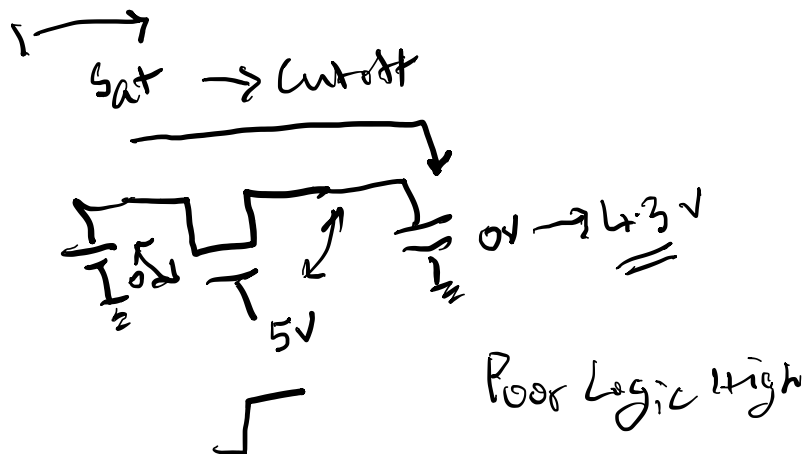
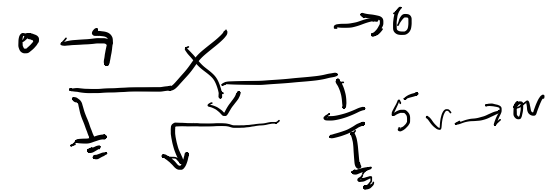
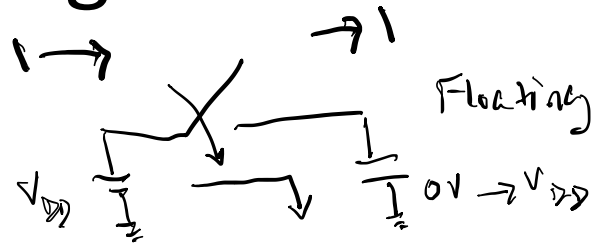
$V_{n3} = 5V$ weak high
good high

$V_{n3} = 4.9V$ poor high

$V_{n3} = 0V$ good low weak low

$V_{n3} = 0.1V$ poor low

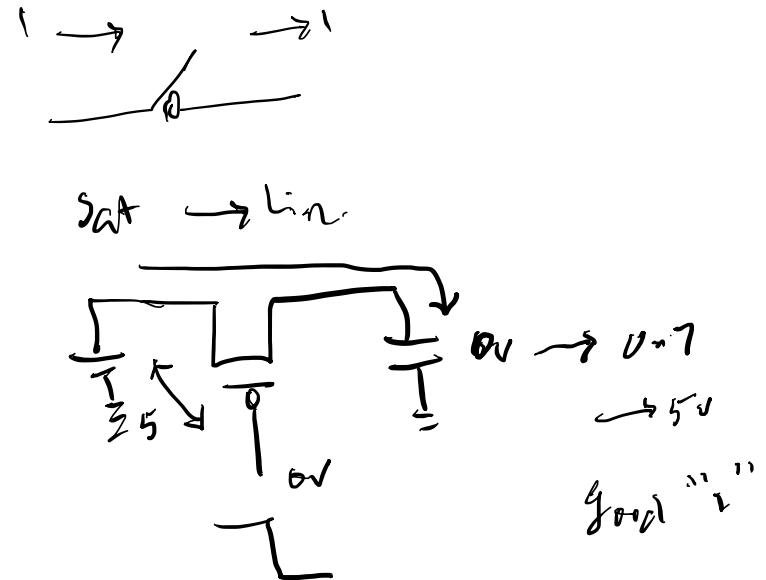
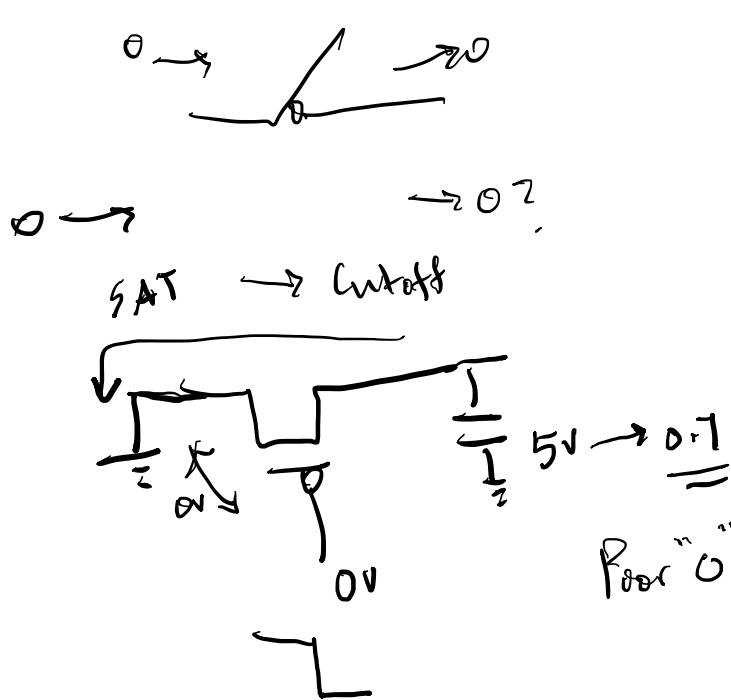
Signal Transmission Through n-Transistor



n-transistor

- Good conductor of logic “0”
- Poor conductor of logic “1”

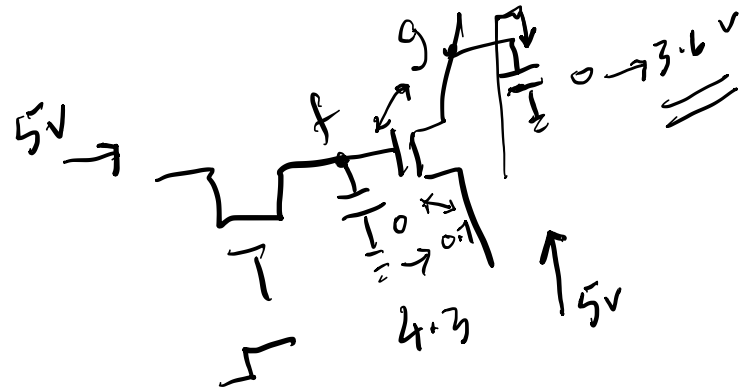
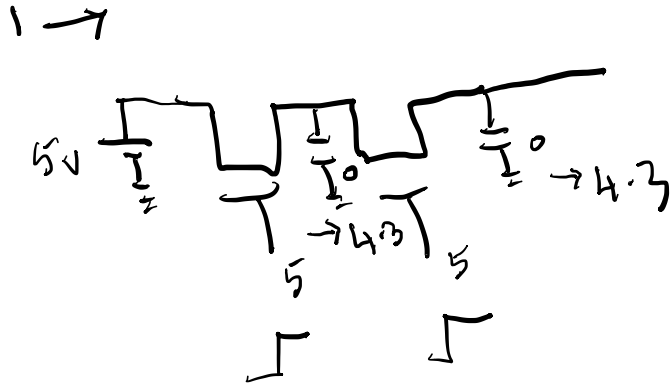
Signal Transmission Through p-Transistor



p-transistor

- Good conductor of logic “1”
- Poor conductor of logic “0”

Pass Transistor Logic – Transmission Quality



X Resistive Drop

✓ Threshold drop